



IoT evolution with Sub 1GHz Wi-Fi SoC

Newracom's NRC7292 represents the first in the industry's line of system-on-chip (SoC) solutions, fully aligned with the IEEE 802.11ah Wi-Fi HaLow specifications. The design is suited for the Internet of Things (IoT) devices. Operating within the Sub



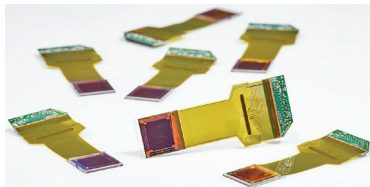
1GHz licence-exempt spectrum, this technology surpasses the capabilities of 2.4GHz and 5GHz technologies, providing an enhanced range and more effective material penetration. The solution minimises the PCB size, requires minimal external circuitries, and enables fast time-to-market. The radio frequency transceiver boasts fully integrated pre-power amplifiers and offers the flexibility to work seamlessly with a variety of commercial external front-end module (FEM) devices.

Newracom

<https://newracom.com>

World's first commercial optical beam steering semiconductor

Lumotive's LM10 is a light control metasurface (LCM) technology that revolutionises 3D sensing. This innovation outperforms traditional lidar



sensors in cost, size, and reliability by eliminating mechanical components. It paves the way for advancements in 3D sensing and various applications dependent on software-controlled optical beamforming. Primarily designed for mid- and short-range tasks, it impacts industries from object tracking to autonomous navigation. The technology's performance, compact size, and software-defined

scanning modes enhance object detection and shape recognition, improving industrial robotics safety and productivity. The features of LCM transform subject tracking and autofocus in photo and video cameras, opening creative opportunities for filmmakers, VR/AR developers, and visual storytellers.

Lumotive

<https://lumotive.com>

Energy-efficient chip transforming AI applications and security

Kneron's KL730, an auto-grade neural processing unit (NPU) chip with an integrated image signal processor (ISP), has a wide range of applica-



tions, from enterprise-edge servers to smart home appliances and advanced driving assistance

systems, with secure and low-energy AI capabilities. KL730 is a first-of-its-kind equipped with a peripheral interface that effortlessly integrates diverse digital signals including image, video, audio, and millimetre waves. The chip is low-cost and achieves a three to fourfold increase in energy efficiency compared to earlier Kneron iterations and is 150 to 200 percent more energy-conservative than its major industry counterparts. The KL730 stands distinctively equipped to revolutionise security within the AIoT domain, facilitating either partial or complete offline operation of GPT models for users.

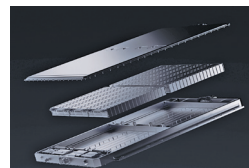
Kneron

<https://www.kneron.com>

World's first 4C superfast charging LFP battery

CATL has launched Shenxing, the world's first 4C superfast charging lithium iron phosphate (LFP) battery, capable of delivering 400 km of

driving range with a 10-minute charge as well as a range of over 700km on a single full charge. The 4C superfast charging, alongside smart algorithm



applications and structural innovations, facilitates a robust driving range, rapid

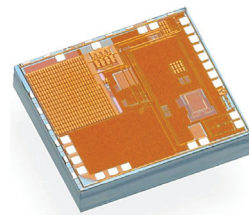
charging even at low temperatures, and an unprecedented level of safety. Additionally, it guarantees consistent performance at low temperatures, capable of reaching up to 80% state of charge in just 10 minutes at room temperature and maintains a rapid 30-minute charging time at temperatures as low as -10°C without compromising acceleration efficiency.

CATL

<https://www.catl.com/en>

Industry's first battery-free IoT metal sensor devices

Identiv, in collaboration with French specialist Asygn, has launched the industry's first sensory tag on metal label, leveraging Asygn's advanced



AS321X IC platform. The devices, designed for battery-free operations, specialise in capturing

temperature and strain data near metallic objects, offering an efficient solution for monitoring without the associated costs and maintenance of battery-powered units. Compliant with RAIN RFID standards, the devices are equipped with an analogue interface for integrating various external sensors, facilitating monitoring of various environmental parameters. This integration of RFID engineering solutions and sensing IC platforms stands out in the industry.

Identiv & Asygn

<https://asygn.com>

<https://www.identiv.com>